ZUGFERD-ENTWICKLERTAGE 2026

Syntax Binding of **EN 16931-1:2026**

Semantics is the cargo, syntax the carrier — what the new standard means for UBL, CII and ZUGFeRD.

Slides as PDF

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Editor of the syntax bindings · CEN/TC 434 WG3 · for DIN

AMENDMENT · MY PROPOSAL 1/2

Round only once (#163)

Only what is paid gets rounded — nothing else.

Paid amounts — rounded once

- BT-115 Amount due for payment, in the Minor Unit (EUR: cent)
- BT-114 Rounding amount carries the one rounding
- BT-113, BT-172, BT-179, BT-183 likewise — they change the amount due

Everything else, VAT included: as calculated

- Prices, lines, totals, taxable amount and VAT
- Cent VAT only via a national CIUS (e.g. UStAE 14.5 (20))
- Rounding to the Minor Unit today stays valid (warning)

WHAT CHANGES

- 6.5.14 applies to the whole chain, not just per rule
- Exact validation (decimal128): equal → valid, within tolerance → warning, otherwise error
- Today's tolerances remain — they now separate warning from error
- No requirement on software: neither a maximum nor a minimum of decimal places
- Syntax unchanged: amounts in UBL and CII are xsd:decimal

Status 14 Oct 2026: Update 3 in WG1 · warning/error following Cyrille Sautereau's 3rd option · open: DRR with cent lines (FR), backward compatibility

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CALCULATION CHAIN · EN 16931-1:2017

From gross price via net price to VAT

One intermediate rounding: **BT-131 to 2 decimal places**

INPUT & INTERMEDIATE VALUES

BT-146	**Item net price** = BT-148 − BT-147	unlimited unrounded	± 0 stated exactly
BT-131	**Invoice line net amount** = BT-146 × BT-129 ÷ BT-149 + Σ BT-141 − Σ BT-136	2 dp interim rounding	± €0.005 per line
BT-106	**Sum of invoice line net amount** = Σ BT-131	2 dp Σ of rounded values	± €0.50 × 100 lines

FINAL RESULTS

BT-116	**VAT category taxable amount** = Σ BT-131 + Σ BT-99 − Σ BT-92	2 dp interim rounding	± €0.50 Σ of rounded lines
BT-117	**VAT category tax amount** = BT-116 × BT-119 ÷ 100	2 dp interim rounding	± €0.10 €0.50 × 19%
BT-110	**Invoice total VAT amount** = Σ BT-117	2 dp Σ of rounded values	± €0.10
BT-115	**Amount due for payment** = BT-110 + BT-109 Invoice total amount without VAT − BT-113 Paid amount + BT-114 Rounding amount	2 dp final value · bank	± €0.60 €0.50 + €0.10

ALLOWED DECIMAL PLACES

BT-148 Item gross price	unlimited
BT-147 Item price discount	unlimited
BT-146 Item net price	unlimited
BT-129 Invoiced quantity	unlimited
BT-149 Item price base quantity	unlimited
BT-141 Invoice line charge or tax amount	2 dp
BT-136 Invoice line allowance amount	2 dp
BT-131 Invoice line net amount	2 dp
BT-106 Sum of invoice line net amount	2 dp
BT-99 Document level charge or tax amount	2 dp
BT-92 Document level allowance amount	2 dp
BT-119 VAT category rate (law)	unlimited

DECIMAL PLACES · FINAL VALUES

BT-116 VAT category taxable amount	2 dp
BT-117 VAT category tax amount	2 dp
BT-110 Invoice total VAT amount	2 dp
BT-115 Amount due for payment	2 dp

Worst case with number of lines and quantity:
→ "How large does the deviation get?"

Red = rounded intermediate value Green = paid amount, rounded once (bank) Orange = rounded input value
dp = decimal places · 2017 (Tab. 26): all amounts fixed 2 dp, regardless of the currency · ± for 100 lines of 100 units, 19% VAT
Rainbow = order of the calculation steps; a BT keeps its colour where it is reused further down

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CALCULATION CHAIN · EN 16931-1:2026

From gross price via net price to VAT

Two intermediate roundings: **BT-146 to Minor Unit + 2** **BT-131 to the Minor Unit**

INPUT & INTERMEDIATE VALUES

BT-146	**Item net price** = BT-148 − BT-147	Minor Unit + 2 interim rounding	± €0.00005 price rounded once
BT-131	**Invoice line net amount** = **BT-146** × BT-129 ÷ BT-149 + Σ BT-141 − Σ BT-136	Minor Unit interim rounding	± €0.01 €0.005 + 100 × €0.00005
BT-106	**Sum of invoice line net amount** = Σ **BT-131**	Minor Unit Σ of rounded values	± €1.00 × 100 lines

FINAL RESULTS

BT-116	**VAT category taxable amount** = Σ **BT-131** + Σ BT-99 − Σ BT-92	Minor Unit interim rounding	± €1.00 Σ of rounded lines
BT-117	**VAT category tax amount** = **BT-116** × BT-119 ÷ 100	Minor Unit interim rounding	± €0.19 €1.00 × 19%
BT-110	**Invoice total VAT amount** = Σ **BT-117**	Minor Unit Σ of rounded values	± €0.19
BT-115	**Amount due for payment** = **BT-110** + **BT-109** Invoice total amount without VAT − **BT-113** Paid amount + **BT-114** Rounding amount	Minor Unit final value · bank	± €1.19 €1.00 + €0.19

ALLOWED DECIMAL PLACES

BT-148 Item gross price	Minor Unit + 2
BT-147 Item price discount	Minor Unit + 2
BT-146 Item net price	Minor Unit + 2
BT-129 Invoiced quantity	4 dp
BT-149 Item price base quantity	2 dp
BT-141 Invoice line charge or tax amount	Minor Unit
BT-136 Invoice line allowance amount	Minor Unit
BT-131 Invoice line net amount	Minor Unit
BT-106 Sum of invoice line net amount	Minor Unit
BT-99 Document level charge or tax amount	Minor Unit
BT-92 Document level allowance amount	Minor Unit
BT-119 VAT category rate (law)	2 dp

DECIMAL PLACES · FINAL VALUES

BT-116 VAT category taxable amount	Minor Unit
BT-117 VAT category tax amount	Minor Unit
BT-110 Invoice total VAT amount	Minor Unit
BT-115 Amount due for payment	Minor Unit

Worst case with number of lines and quantity:
→ "How large does the deviation get?"

Red = rounded intermediate value **Green** = paid amount, rounded once (bank) **Orange** = rounded input value
dp = decimal places · Minor Unit = smallest currency unit, ISO 4217 (EUR: 2 dp) · ± for 100 lines of 100 units, 19% VAT
Rainbow = order of the calculation steps; a BT keeps its colour where it is reused further down

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CALCULATION CHAIN · PROPOSED EN 16931-1:2026 AMENDMENT (#163)

From gross price via net price to VAT

No intermediate rounding — only BT-115 Amount due for payment is rounded; BT-114 Rounding amount shows it.

INPUT & INTERMEDIATE VALUES

BT-146	**Item net price** = BT-148 − BT-147	free unrounded	± 0 as agreed
BT-131	**Invoice line net amount** = **BT-146** × BT-129 ÷ BT-149 + Σ BT-141 − Σ BT-136	free unrounded	± 0 exact
BT-106	**Sum of invoice line net amount** = Σ **BT-131**	free unrounded	± 0 exact sum

FINAL RESULTS

BT-116	**VAT category taxable amount** = Σ **BT-131** + Σ BT-99 − Σ BT-92	free unrounded	± 0 exact
BT-117	**VAT category tax amount** = **BT-116** × BT-119 ÷ 100	free unrounded	± 0 Minor Unit only via CIUS → warning
BT-110	**Invoice total VAT amount** = Σ **BT-117**	free unrounded	± 0
BT-115	**Amount due for payment** = **BT-110** + **BT-109** Invoice total amount without VAT − **BT-113** Paid amount + **BT-114** Rounding amount	Minor Unit rounded once	± 0 BT-114 shows the rounding

ALLOWED DECIMAL PLACES

BT-148 Item gross price	free
BT-147 Item price discount	free
BT-146 Item net price	free
BT-129 Invoiced quantity	free
BT-149 Item price base quantity	free
BT-141 Invoice line charge or tax amount	free
BT-136 Invoice line allowance amount	free
BT-131 Invoice line net amount	free
BT-106 Sum of invoice line net amount	free
BT-99 Document level charge or tax amount	free
BT-92 Document level allowance amount	free
BT-119 VAT category rate (law)	2 dp

DECIMAL PLACES · FINAL VALUES

BT-116 VAT category taxable amount	free
BT-117 VAT category tax amount	free
BT-110 Invoice total VAT amount	free
BT-115 Amount due for payment	Minor Unit

Worst case with number of lines and quantity:
→ "How large does the deviation get?"

Red = rounded intermediate value Green = paid amount, rounded once (bank) Orange = rounded input value
dp = decimal places · Minor Unit = smallest currency unit, ISO 4217 (EUR: 2 dp) · ± for 100 lines of 100 units, 19% VAT
Rainbow = order of the calculation steps; a BT keeps its colour where it is reused further down

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WORST CASE · EN 16931-1:2017

How large does the deviation get?

One factor: the number of lines n — each line rounds **BT-131** Invoice line net amount and loses up to ± €0.005.

Factor 1 · price rounding × quantity Q

BT-146 Item net price is unlimited in 2017 — the price is not rounded, so the quantity magnifies no error.

`none`

Factor 2 · number of lines n

Each line rounds **BT-131** Invoice line net amount to 2 decimal places: up to ± €0.005. **BT-106** Sum of invoice line net amount and **BT-116** VAT category taxable amount add up all n lines — and with them the errors of all lines.

`± n × €0.005`

NET DEVIATION **BT-106** Sum of invoice line net amount / **BT-116** VAT category taxable amount

Units per line (Q) → ↓ Lines (n)	100 units	10,000 units	1,000,000 units
1 line	Price × quantity ± 0 + rounding BT-131 ± €0.005 = ± €0.005	Price × quantity ± 0 + rounding BT-131 ± €0.005 = ± €0.005	Price × quantity ± 0 + rounding BT-131 ± €0.005 = ± €0.005
100 lines	± €0.50	± €0.50	± €0.50
10,000 lines	± €50.00	± €50.00	± €50.00

Assumption: all roundings go in the same direction, BT-149 Item price base quantity = 1, 19% VAT.

TOTAL · NET

n × €0.005

VAT: 19% of that. Open-ended — the more lines, the larger the deviation.

100 lines of 10,000 units
Net ± €0.50 VAT ± €0.10

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WORST CASE · EN 16931-1:2026

How large does the deviation get?

Two factors: first the quantity Q × rounded **BT-146** Item net price, then the number of lines n — open-ended.

Factor 1 · price rounding × quantity Q

BT-146 Item net price is "squeezed" once into Minor Unit + 2 (EUR: 4 dp): up to ± €0.00005. **BT-131** Invoice line net amount multiplies this one error by **BT-129** Invoiced quantity. #148: €0.123449 → €0.1234; with 10,000 units €0.49 is missing.

`± Q × €0.00005 per line`

Factor 2 · number of lines n

Each line rounds **BT-131** Invoice line net amount additionally to the Minor Unit: up to ± €0.005. **BT-106** Sum of invoice line net amount and **BT-116** VAT category taxable amount add up all n lines — and with them the errors of all lines.

`± n × €0.005`

NET DEVIATION **BT-106** Sum of invoice line net amount / **BT-116** VAT category taxable amount

Units per line (Q) → ↓ Lines (n)	100 units	10,000 units	1,000,000 units
1 line	Price × quantity ± €0.005 + rounding BT-131 ± €0.005 = ± €0.01	Price × quantity ± €0.50 + rounding BT-131 ± €0.005 = ± €0.505	Price × quantity ± €50.00 + rounding BT-131 ± €0.005 = ± €50.005
100 lines	± €1.00	± €50.50	± €5,000.50
10,000 lines	± €100.00	± €5,050.00	± €500,050.00

Assumption: all roundings go in the same direction, BT-149 Item price base quantity = 1, 19% VAT. Likewise: BT-129 Invoiced quantity only 4 dp — up to ± 0.00005 × price per line for expensive items.

TOTAL · NET

n × (Q × €0.00005 + €0.005)

VAT: 19% of that. Open-ended — more lines or larger quantities, more deviation.

100 lines of 10,000 units
Net **± €50.50** VAT **± €9.60**

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WORST CASE · PROPOSED AMENDMENT (#163)

How large does the deviation get?

No factor: without intermediate rounding it stays at ± 0 — for any quantity and any number of lines.

Factor 1 · price rounding × quantity Q

BT-146 Item net price and **BT-131** Invoice line net amount stay exact — the quantity magnifies no error.

none

Factor 2 · number of lines n

BT-131 Invoice line net amount stays exact — it does not matter how many lines **BT-106** Sum of invoice line net amount and **BT-116** VAT category taxable amount add up.

none

NET DEVIATION **BT-106** Sum of invoice line net amount / **BT-116** VAT category taxable amount

Units per line (Q) → ↓ Lines (n)	100 units	10,000 units	1,000,000 units
1 line	Price × quantity ± 0 + rounding BT-131 ± 0 = ± 0	Price × quantity ± 0 + rounding BT-131 ± 0 = ± 0	Price × quantity ± 0 + rounding BT-131 ± 0 = ± 0
100 lines	± 0	± 0	± 0
10,000 lines	± 0	± 0	± 0

Assumption: all roundings go in the same direction, BT-149 Item price base quantity = 1, 19% VAT.

TOTAL · NET

± 0

Only BT-115 Amount due for payment is rounded once; BT-114 Rounding amount shows it.

100 lines of 10,000 units
Net ± 0 VAT ± 0

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AMENDMENT · MY PROPOSAL 2/2

Gross prices for the EN core (#164)

If the price is fixed including VAT, it should lead the invoice.

Step	Net model (today)	Gross model (new)
Price	BT-146 Item net price	BT-146V Item net price plus VAT
Line	BT-131 = price × quantity …	BT-131V = price plus VAT × quantity …
Per category	BT-116 = Σ BT-131 …	BT-116V = Σ BT-131V …
VAT	BT-117 = BT-116 × rate added	BT-117 = BT-116V × rate ÷ (100 + rate) extracted
Net per category	(given)	BT-116 = BT-116V − BT-117

The model is selected via BT-146V on each line — no indicator needed. BT-146 remains mandatory (Art. 226(8) VAT Directive), in the gross model for information only. Not for categories N, D, F, I, J.

Often impossible today: At 19% VAT, 16% of all gross amounts cannot be reached with net and VAT in the Minor Unit — €9.99 = 8.39 + 1.59 → 9.98. With 5 lines 52%, with 35 lines 81% miss the receipt total.

EXAMPLE: FIXED BOOK PRICES

1,000 books at €24.95 incl. 7% VAT

	Total	VAT
Target	24,950.00	1,632.24
Net 2026	24,950.05	1,632.25
Gross model	24,950.00	1,632.24

Net model: 5 cents too much, VAT 1 cent too high. Gross model: VAT €1,632.24299… → €1,632.24 only due to the cent rule (UStAE) — #163: warning.

Use cases: **fixed book prices · receipt → invoice · retail and marketplaces · statutory gross prices**

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CONTACT

Talk to me

Slides as PPTX

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Syntax bindings · validation · tooling
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Follow & join in

- WG1 issues #163 and #164
- Issues on the syntax bindings (WG3)
- Svanton: exact arithmetic for Saxon
- XSLT 4.0 proposal (qt4cg #2935)

Thank you for your attention — I look forward to the exchange at the ZUGFeRD-Entwicklertage.

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